

Task 2: Pastel Watercolor Expense Tracker

1. Project Overview

A visually stunning, traditional Python desktop application built with Tkinter and Matplotlib. This expense tracker brings a soft, calming pastel watercolor aesthetic to personal finance management, complete with gradient backgrounds, colorful icons, and dynamic charts.

2. Features

- Aesthetic Desktop UI with programmatically generated pastel gradients
- Interactive Matplotlib charts (Category Breakdown Bar Chart, Expense Trend Line Graph)
- Colorful CLI startup with vibrant ASCII art using Colorama
- Data persistence via local JSON file
- Budget tracking with instant comparison against a customizable limit

3. Challenges & Solutions

- Challenge: Embedding Matplotlib charts seamlessly into a Tkinter window.
Solution: Used FigureCanvasTkAgg to bind the charts into Tkinter frames and applied custom pastel themes to match the overall UI.
- Challenge: Generating custom watercolor aesthetics programmatically.
Solution: Used Pillow (PIL) to create gradients and rounded rectangles that Tkinter natively lacks.

4. What I Learned

- Learned how to integrate Matplotlib for data visualization within a Tkinter desktop app.
- Enhanced understanding of true-color ANSI escape sequences via Colorama.
- Mastered programmatic image generation using Pillow for advanced UI customization.